

UU1231 | *Rocky Mountain University*

Building An Automated Registrar Workflow

Capstone Project Report | Summer 2026

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Executive Summary

Federal regulations and university policies require higher education institutions such as Rocky Mountain University of Health Professions to track and provide evidence of student participation in enrolled courses. The Registrar's Office historically managed the auditing of this process manually, which required staff to spend several hours each week pulling Canvas logs, cleaning spreadsheets, cross-referencing activity across multiple systems, and sending individual warning emails to students. This manual process takes significant time and leaves room for human error.

To help the Registrar's Office manage its time more efficiently, our capstone team analyzed, designed, and built an automated attendance-tracking pipeline using the Microsoft Power Platform. The system extracts attendance information from a data Lakehouse table, runs automated validation logic, and flags inactive students in a central SharePoint list. Registrar staff then review the SharePoint records and approve them before automated Power Automate flows send standardized warning emails, log actions in the Student Information System, and update the record in the SharePoint list. This solution removes hours of manual data work, improves compliance tracking, and helps ensure students receive timely support.

Project Overview

Our team divided responsibilities to balance individual opportunities for growth and leverage strengths. Sooyoung Lee managed the project and led frontline user education. Paul Arnold gathered requirements and directed user acceptance testing. Robert Bost designed and implemented updates to the workflow automations. Throughout the project, we worked closely with RMU IT sponsors Dan Harmuth, Jerry Van Wie, and Daniel Vela, as well as multiple staff members from the Registrar's Office.

For deeper context, Rocky Mountain University of Health Professions delivers graduate-level healthcare education. To maintain federal funding and institutional standing, the university must track student attendance weekly. However, as the student body has grown, the manual process has created operational bottlenecks. Each week, staff received raw Canvas CSV files that required tedious cleaning. Staff had to reformat dates for sorting, fill in missing section numbers in the Student Information System, mark full- and partial-semester courses, and check exclusion rules for holidays, on-site sessions, and non-degree students. Additionally, the CSV only checked whether certain activities were performed on Canvas, such as submitting an assignment or a quiz. Once the CSV was scrubbed, the Registrar's Office staff would manually check Canvas activity logs for other online activities that count as participation, such as discussion posts, messages, or downloads. If the student had not performed any tasks that qualify as participation, staff coordinated with the Student Affairs office via chat to confirm known student exceptions that may preclude participation, such as bereavement, medical leave, or parental leave. Once it was confirmed that the student did not meet the attendance criteria, staff manually drafted a first- or second-warning email. This workflow was complex, error-prone, and time-consuming.

Results & Planning Analysis

The project delivered all outcomes outlined in our Statement of Work. We documented the current and future-state workflows and designed a data pipeline linking the data warehouse to SharePoint and Power Automate. We created a SharePoint review portal for Registrar staff, and delivered technical documentation and training materials. A full list of the deliverables agreed

upon in the statement of work is shown in image 1 below.

Deliverables

SOW Deliverable	Project Outcome
Workflow Analysis	Documented As-Is and To-Be registrar workflows
System Architecture	Designed data flow between Data Warehouse, SharePoint, and Power Automate
Automation Pipeline	Developed an automated attendance processing workflow
SharePoint Solution	Built review and approval interface for Registrar staff
Testing & Validation	Compared automated results against the manual process
Documentation & Training	Created technical documentation and user guidance

Image 1: Deliverables

To better understand the process being developed and to ensure data requirements were met, we began our analysis by observing it in the Registrar's office. After observing the process, we mapped out the current workflow and collaborated with front-line and technical staff to develop the proposed future workflow. Please see images 2 & 3 below and pay particular attention to the swim lane titled "Registrar's Office." You can see in image 2 that employees were managing several repetitive tasks. In the future state, the steps that required human judgment became the only processes that remained in their hands. The other steps were essentially outsourced to the automated flow.

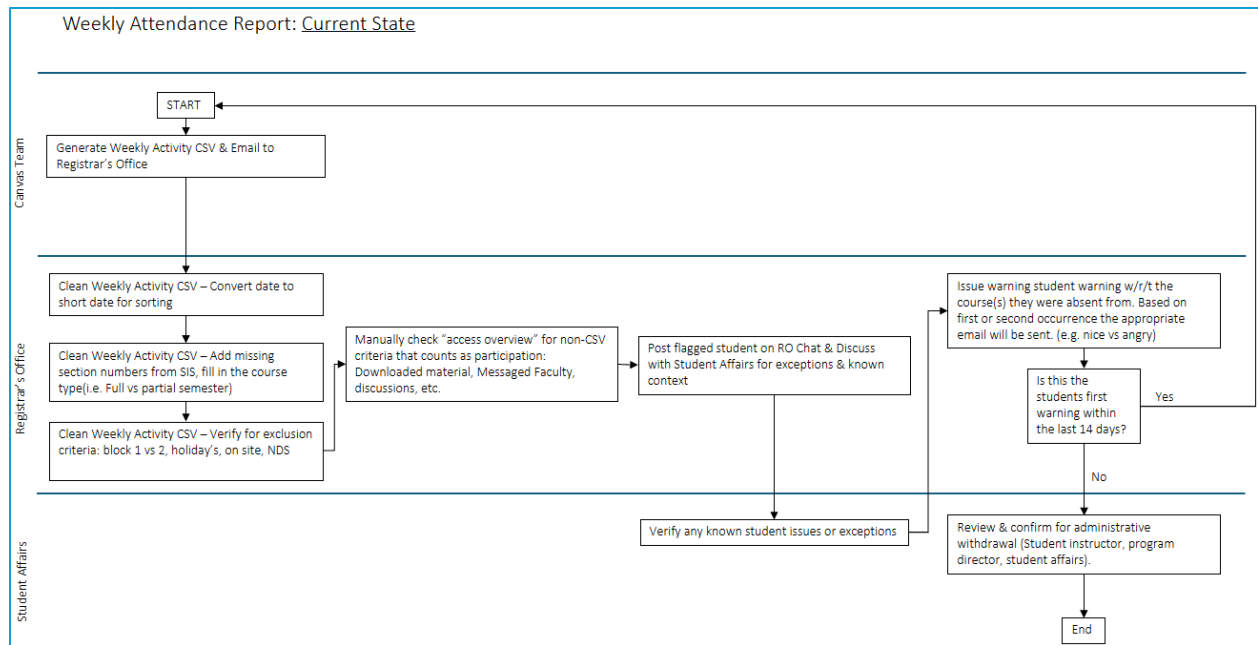


Image 2: Current State

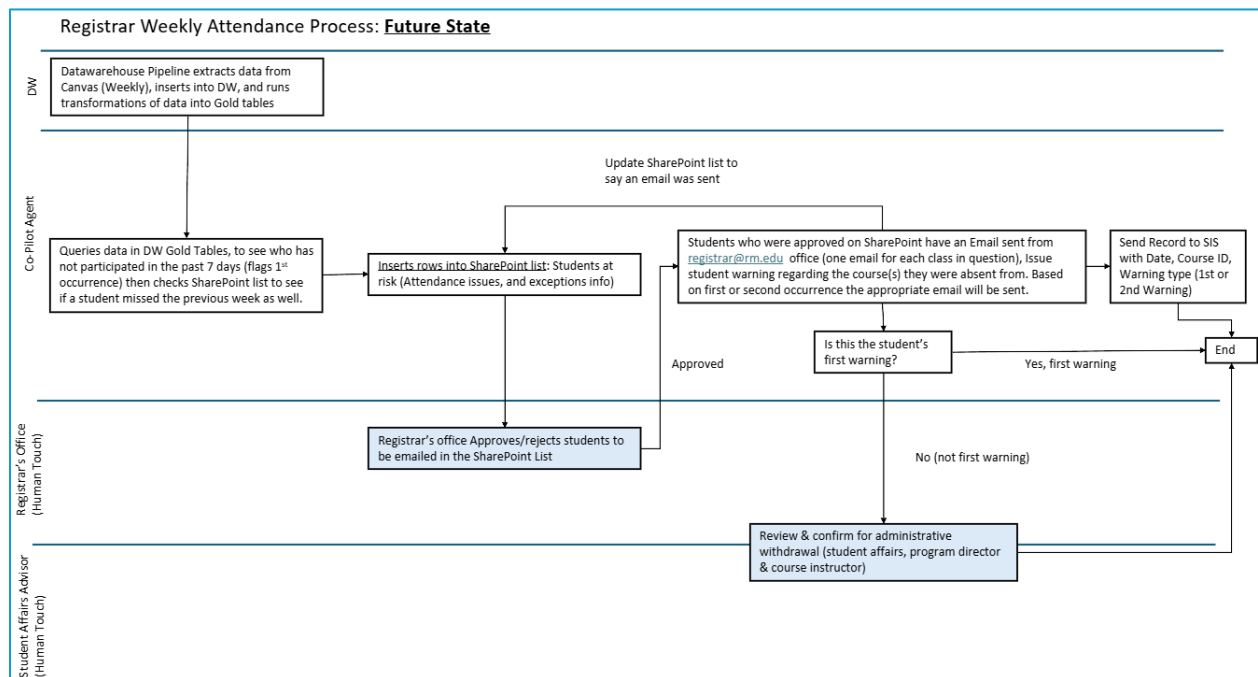


Image 3: Future State

Design & Implementation

The new system simplifies the weekly review cycle. Initially, we collaborated with data engineering to modify the CSV and produced a gold table that handled all the manual data cleaning the registrar's office was performing. Additionally, a database administrator was included to identify the tables for items in Canvas that the staff were manually searching each week. Once these tables were cleaned and validated, we began developing the automated power application.

Using Microsoft Power Apps, the gold tables were called and queried to filter students who were out of compliance with the attendance policy. It checks whether any students exceeded the seven-day activity thresholds, evaluates warning history, applies course exclusion rules, and inserts flagged records into a central SharePoint list that the RO can view (see image 4).

Student ID	First Name	Last Name	Email	Course & Section	Last Participation Original	Cut Off Date	Days Since L...	Participation Flag	Status
				DMSc-HLA 670-1 Organizational Behavior & Management in Healthcare	2026-07-15 13:14:56.760000	07/22/2026 13:14:56	9	No	Pending Review
				SLPD-SLP 711A+ Capstone Residency	2026-07-12 19:12:08.678000	07/19/2026 19:12:08	12	No	Pending Review
				SLPD-SLP 711A+ Capstone Residency	2026-07-12 22:14:09.709000	07/19/2026 22:14:09	12	No	Pending Review
				PHD-HS 833 B-1 Proposal Writing	2026-07-13 09:22:51.922000	07/20/2026 09:22:51	11	No	Pending Review
				PHD-HS 833 B-1 Proposal Writing	2026-07-15 21:43:50.419000	07/22/2026 21:43:50	9	No	Pending Review
				DMSc-HLA 670-1 Organizational Behavior & Management in Healthcare	2026-07-15 13:52:04.832000	07/22/2026 13:52:04	9	No	Pending Review
				NURS-DNP 715-1 Clinical Leadership II: Clinical Focus	2026-07-14 16:23:54.319000	07/21/2026 16:23:54	10	No	Pending Review

Image 4

Registrar staff then log into this SharePoint list to review flagged records. The interface displays student names, IDs, course sections, last activity dates, days inactive, and current review status. Staff approve or reject each record. When staff approves a record, a Power Automate flow is

triggered to send a standardized email template with the appropriate warning, update the record's status, and create a record for the Registrar's Office for future audits.

End-to-End Flow



Image 5: End-to-End Flow

Lessons Learned

Building this automation taught us important lessons about process design, stakeholder management, and technology.

From a process design perspective, we learned the importance of gathering user requirements to understand client requests and to evaluate what we were technically capable of delivering. We met regularly, and as the project progressed, our meeting frequency increased. This helped ensure we stayed on track, satisfied user needs, and met stakeholder requirements. Midway through the project, additional requirements surfaced that were not part of the original statement of work. After discussing the importance of fidelity to the SOW and the implications of scope creep, we reached a practical compromise by adapting a modular code snippet provided by one of the RMU stakeholders.

From a personal perspective, the project showed us that automation works best when it supports frontline staff rather than replacing human decision-making. Managerially, frequent feedback loops produced better adoption than delivering a finished system at the very end.

From a technical perspective, we learned that even seemingly simple tasks, like the ones the Registrar's Office performs, still require a clear architecture, detailed error handling, and disciplined testing to run reliably in production environments. Overall, the project strengthened our discipline in requirements gathering, deepened our empathy for end users, increased our skills in enterprise low-code development, and improved our ability to communicate complex technical ideas clearly.

Next Steps

To ensure a smooth transition after we finished the capstone project, we involved an analyst and a technical project manager from RMU's staff throughout the assignment. And while there is likely still improvement and iteration to be done on the program we developed, we also recommend a handful of next steps:

- Run the automated flow in parallel with the manual process for several weeks to confirm accuracy and track time savings.
- Conduct hands-on training sessions for all Registrar staff using the SharePoint review list.
- Expand the shared exception form to cover additional edge cases discovered during parallel testing.

- Connect the SharePoint history to Power BI, giving leadership clear visibility into weekly compliance trends.
- Create a technical handover package including runbooks and flow diagrams so RMU IT can own and maintain the system moving forward.

Conclusion

Building this solution demonstrated how effective systems analysis and design, project management, and technical skills can mesh. Working together with multiple teams across the Rocky Mountain Universities' faculty was a rewarding way to learn, but the true measure of success lies beyond the work we did. We hope this automation provides tangible benefits to the frontline staff we serve, lifting the weight of repetitive tasks so they can focus their energy where it matters most: supporting students.